

Module 5

Thrashing

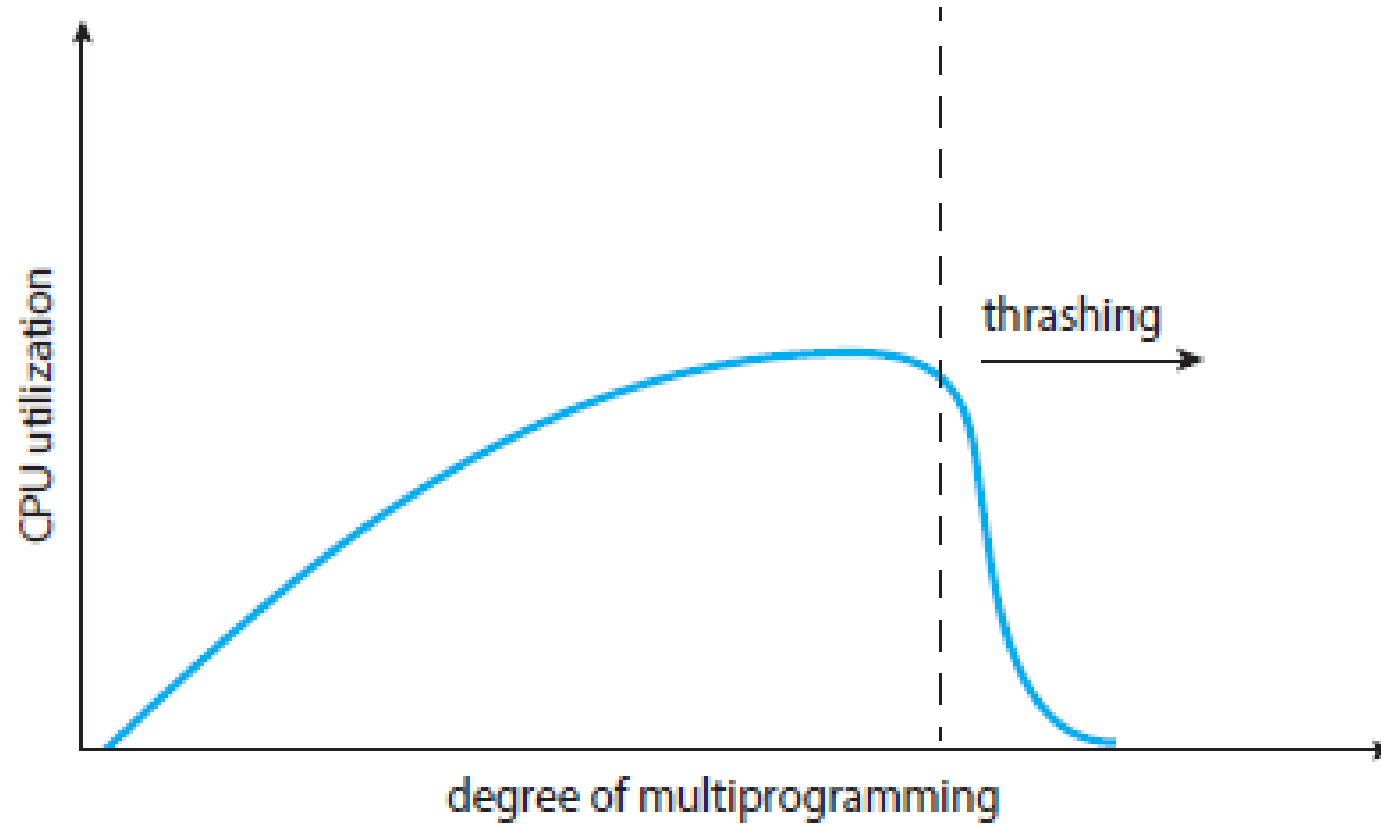
Thrashing

- In multiprogramming, there can be a scenario when the system spends most of its time transferring pages between the main memory and the secondary memory due to frequent page faults.
- A process is said to be thrashing if the CPU spends more time serving page faults than executing the pages.
- This leads to low CPU utilization and the Operating System in return tries to increase the degree of multiprogramming.

Thrashing - Causes

- High degree of Multiprogramming.
- Less number of frames compared to the processes required.
- The process scheduling scheme which swaps in more processes when CPU utilization is low.

Thrashing - Causes



Thrashing - Algorithms

Global Page Replacement

- Global allocation allows a process to select a replacement page from the set of all frames. This implies that if a process is thrashing, it can steal frames from other processes and cause them to thrash too.
- This scheme can increase the effect of thrashing.

Thrashing - Algorithms

Local Page Replacement

- Local allocation requires each process to select a replacement page from its own set of allocated frames. This scheme can reduce thrashing to some extent by restricting them from acquiring global frames.
- Because the thrashing processes are in the waiting queue for most of the time clogging the queue, the **average time to serve a page fault increases**. This in turn increases the average time to process an IO operation, affecting not-thrashed processes too.
- if one process is not using all the assigned frames, other processes also cannot occupy it. These wasted frames could have otherwise helped a high paging process to come out of thrashing

Thrashing - Elimination Technique

- **Adjust the swap file size:** The swap file is a portion of the hard disk that is used as virtual memory. If the swap file is too small, it can cause thrashing. By increasing the size of the swap file, the system will have more virtual memory available, which can help reduce thrashing.
- **Increase the amount of RAM:** If the system does not have enough physical memory to meet the demands of running processes or applications, adding more RAM can help reduce thrashing.

Thrashing - Elimination Technique

- **Decrease the number of applications running on the computer:**
Running too many applications at once can cause thrashing by placing a heavy demand on the system's memory resources. Closing unnecessary applications can help free up memory and reduce the likelihood of thrashing.
- **Replace programs:** Some programs are more memory-intensive than others. If a particular program is causing thrashing, replacing it with a more efficient alternative may help alleviate the problem.

Thrashing - Locality Model

- To prevent thrashing, we must *provide a process with as many frames as it needs*.
- **But how do we know how many frames it “needs”?**
- One strategy starts by looking at how many frames a process is actually using. This approach defines the **locality model** of process execution.
- **A locality is a set of pages that are actively used together.**
- A running program is generally composed of several different localities, which may overlap.

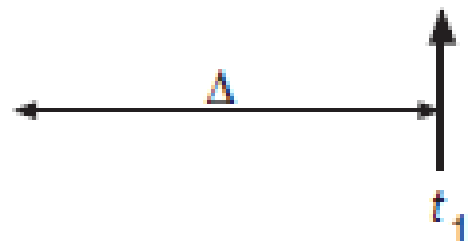
Thrashing – Working Set Model

- The **working-set model** is based on the assumption of locality. This model uses a parameter, Δ , to define the **working-set window**.
- The idea is to **examine the most recent Δ page references**. The set of pages in the most recent Δ page references is the **working set**.
- If a page is in active use, it will be in the working set. If it is no longer being used, it will drop from the working set Δ time units after its last reference.
- The most important property of the working set, then, is its **size**

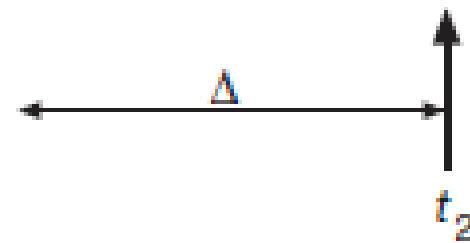
Thrashing - Working Set Model

page reference table

... 2 6 1 5 7 7 7 7 5 1 6 2 3 4 1 2 3 4 4 4 4 3 4 3 4 4 4 4 1 3 2 3 4 4 4 4 3 4 4 4 ...



$$WS(t_1) = \{1, 2, 5, 6, 7\}$$



$$WS(t_2) = \{3, 4\}$$

Thrashing - Page-fault frequency

- Thrashing has a high page-fault rate.
- To control the page-fault rate, when it is **too high**, the process **needs more frames**.
- Conversely, if the page fault rate is **too low**, then the process may **have too many frames**.
- Establish upper and lower bounds on the desired page-fault rate
- If the actual page-fault rate exceeds the upper limit, we allocate the process another frame. If the page-fault rate falls below the lower limit, we remove a frame from the process.